ELEC5208 Active Distribution Systems

Group Assignment

TASK I Report: Distribution Network Modelling and Power Flow

Group members:

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1 Introduction

With the increasing prevalence of distributed energy resources (DERs), such as rooftop solar PV and home energy storage systems, distribution networks face new operational challenges, including bidirectional power flow, voltage rise, and line overload.

This task aims to build a three-phase four-wire low-voltage distribution network model using OpenDSS and incorporate real smart meter data to simulate user loads. Based on this, a year-round time-series power flow analysis is performed to assess voltage variations and network performance.

2 Network Modelling

2.1 Network Topology

A radial three-phase four-wire LV network was constructed in OpenDSS. The main feeder was arranged as:

Tbus → B1 → B2 → B3 → B4 → B5 → B6

with three branch buses:

- B2a branching from B2
- B4a branching from B4
- B6a branching from B6

This topology was selected for three reasons.

First, it satisfies the assignment requirement that the network must be radial. In a radial network, each downstream bus has only one electrical path back to the source, which simplifies protection, interpretation of voltage drop, and power flow tracing.

Second, the topology ensures that there are at least five line segments between the transformer and the furthest bus. In the present model, the furthest path is:

Tbus → B1 → B2 → B3 → B4 → B5 → B6 → B6a

which contains seven feeder sections after the transformer secondary bus. Therefore, the requirement is comfortably satisfied.

Third, the chosen topology provides a meaningful “electrically remote” end bus. This is important because one of the key aims of the task is to examine how voltage varies with location. If the feeder were too short or too compact, the voltage drop behaviour would be weak, and the results would not be very informative.

The line lengths used in the model are:

- Tbus–B1: 0.08 km
- B1–B2: 0.08 km
- B2–B3: 0.07 km
- B3–B4: 0.07 km
- B4–B5: 0.05 km
- B5–B6: 0.04 km
- B2–B2a: 0.06 km
- B4–B4a: 0.06 km
- B6–B6a: 0.05 km

Hence, the total network length is:

$$\mathbf{0.08 + 0.08 + 0.07 + 0.07 + 0.05 + 0.04 + 0.06 + 0.06 + 0.05 = 0.56\ km}$$

Thus, the designed network length is 0.56 km, which satisfies the assignment requirement that the total feeder length must lie between 0.3 km and 1.5 km.

The main advantage of this topology is that it is simple enough to interpret while still being realistic enough to generate meaningful voltage gradients and branch effects. If a much shorter network had been used, the voltage drop would likely have been too small to study properly. Conversely, if the network had been made excessively long or heavily branched, it could have introduced unnecessary modelling complexity and increased the likelihood of poor voltage performance.

2.2 Transformer and Grounding

The feeder is supplied by a 500 kVA, 11 kV / 0.4 kV distribution transformer configured as delta-grounded-wye. This directly satisfies the assignment requirement that the transformer rating should be between 150 kVA and 500 kVA, with a nominal voltage ratio of 11 kV / 400 V line-to-line.

The choice of 500 kVA has both strengths and trade-offs. The main advantage is that it provides sufficient headroom for the aggregated residential load, ensuring that the transformer does not become the dominant bottleneck during peak demand. This allows the study to focus more clearly on feeder voltage behaviour rather than transformer overload. A smaller transformer might have produced more severe voltage stress and higher loading sensitivity, but it could also have pushed the model closer to thermal or numerical limits, making interpretation less clean.

On the LV side, the transformer neutral is grounded through a 0.3 Ω resistor, while each load bus neutral is grounded through a 10 Ω resistor. These values were taken directly from the assignment requirements. Including explicit grounding resistors is important in a four-wire LV model because the neutral conductor and grounding arrangements affect phase-to-neutral voltages and unbalance behaviour. If grounding had been omitted or treated too simplistically, the resulting phase voltages could have been less realistic, particularly under unbalanced single-phase loading.

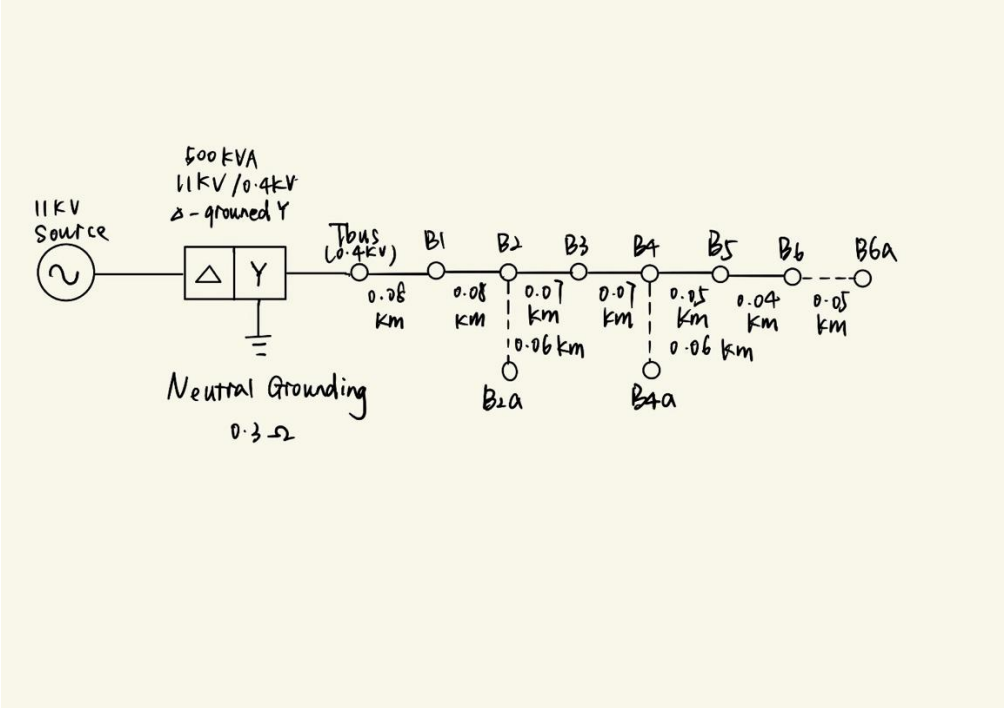


Figure 1 Network Topology & Transformer & Grounding Diagram

2.3 Conductor and Line Modelling

The conductor selected for all feeder sections is Neptune, taken from the candidate conductor list provided in the assignment. Its parameters are:

- Diameter = 16.3 mm
- AC resistance = 0.224 Ω/km

- NormAmps = 479 A

This conductor was chosen because it provides a good balance between relatively low AC resistance and a high thermal current rating. For the present study, this is advantageous because it reduces the risk that excessive feeder resistance alone dominates the voltage drop, while still representing a realistic LV overhead conductor.

The line geometry was modelled according to the assignment specification for a four-wire open-wire overhead arrangement. The conductor coordinates are:

(−110,580), (−45,580), (45,580), (110,580) cm

which correspond to the required phase and neutral spacing. This geometry was explicitly implemented in OpenDSS using WireData, LineSpacing, and LineGeometry. This is a better modelling choice than assigning an arbitrary lumped line impedance, because it ensures the feeder impedance representation is consistent with the physical conductor arrangement required by the task.

If a conductor with significantly higher resistance had been chosen, the voltage drop would have increased, particularly at remote buses such as b6 and b6a. On the other hand, choosing an unrealistically strong conductor simply to suppress voltage drop would weaken the engineering credibility of the model. Therefore, the selected Neptune conductor offers a reasonable compromise between realism and network performance.

In this study, the line impedance modelling was implemented using the OpenDSS built-in functionality based on physical conductor data and geometric configuration, rather than manually defining lumped impedance values. Specifically, the combination of WireData, LineSpacing, and LineGeometry allows OpenDSS to internally construct the multi-conductor impedance matrix corresponding to the four-wire overhead line.

This approach ensures that the electrical characteristics of the feeder are consistent with the physical conductor arrangement specified in the assignment. Compared with simplified impedance assignment, this method better captures the coupling effects between phase conductors and the neutral, which is particularly important in low-voltage four-wire systems where unbalance and neutral current can significantly influence voltage behaviour.

3 Smart Meter Data Processing and Load Construction

3.1 Raw data and data interpretation

The smart meter data were provided by SolarAnalytics at a 5-minute resolution. Each original CSV contains the fields:

- utc_timestamp
- site_id
- pv_generation
- total_load
- localtime

These variables represent real customer-side measurements over a full year. The pv_generation field corresponds to PV output, while total_load represents the measured net load behaviour at the site.

A key issue encountered in the raw data was the presence of negative values. In practice, negative PV generation is physically unrealistic and is most likely associated with sensor offset, metering noise, or data artefacts. Negative total load values may represent export conditions or data anomalies. Since the present Task 1 focuses on modelling customer demand in the LV network rather than explicitly modelling distributed generation injection, negative values in both series were clipped to zero.

This data treatment has both strengths and limitations. Its main advantage is that it prevents the load model from unintentionally becoming a generation model in OpenDSS. If negative total_load values had been used directly, some loads could effectively behave as embedded generation, which would fundamentally change the interpretation of the feeder study. However, the limitation is that this treatment removes explicit export behaviour and therefore under-represents scenarios where local PV could reduce feeder demand or raise voltages.

3.2 Profile standardisation and missing data handling

A complete year with 5-minute resolution should contain:

$$\mathbf{365 \times 24 \times 12 = 105120}$$

time points. During processing, it was found that not all original profiles contained exactly 105120 entries. This indicates that some source profiles were incomplete, most likely due to missing intervals in the original data rather than errors in the simulation code itself.

To ensure that all yearly load profiles were compatible with a common full-year simulation timeline, every processed profile was forced to a length of 105120 points. If a profile was longer,

it was truncated to the first 105120 points. If a profile was shorter, the missing tail portion was padded using the final available value.

The reason for this step is methodological consistency. OpenDSS LoadShape objects must align with the simulation time horizon if the resulting feeder power and voltage traces are to be meaningfully compared across all customers. If profile lengths are inconsistent, some loads effectively terminate earlier than others, which makes the yearly simulation logically inconsistent.

The advantage of the chosen method is that it preserves all existing valid data while enforcing a common simulation length. The limitation is that padded tail values are not as physically informative as measured values. However, this is still preferable to leaving the profiles inconsistent, because an inconsistent yearly timeline would undermine the validity of the full-year study.

3.3 Load profile allocation and scaling

In the final model, 9 distinct smart-meter profiles were selected from the available dataset and expanded to 36 single-phase customers through repeated allocation with different scaling factors, bus locations, and phase assignments.

A total of 36 single-phase loads were created, satisfying the requirement of more than 30 loads, with at least 10 connected to each phase. In this model, the loads were distributed approximately evenly across the three phases, i.e. 12 loads per phase, to maintain reasonable phase balance while still allowing time-varying unbalance due to different customer profiles.

The 36 loads were mapped from a smaller set of selected real household profiles using a load allocation table. Each entry specifies:

- load ID
- bus location
- phase connection
- assigned profile file
- scaling factor

This approach has several advantages. First, it preserves diversity in load behaviour because different loads can follow different real customer patterns. Second, the scaling factors prevent the network from becoming unrealistically uniform.

If all 36 loads had been identical, the feeder behaviour would have become overly synchronous and less realistic. On the other hand, if too many large profiles had been assigned to the remote buses, the feeder could have become unnecessarily stressed, producing excessive voltage drop or even numerical convergence issues. Therefore, the final allocation represents a balance between realism, diversity, and numerical stability.

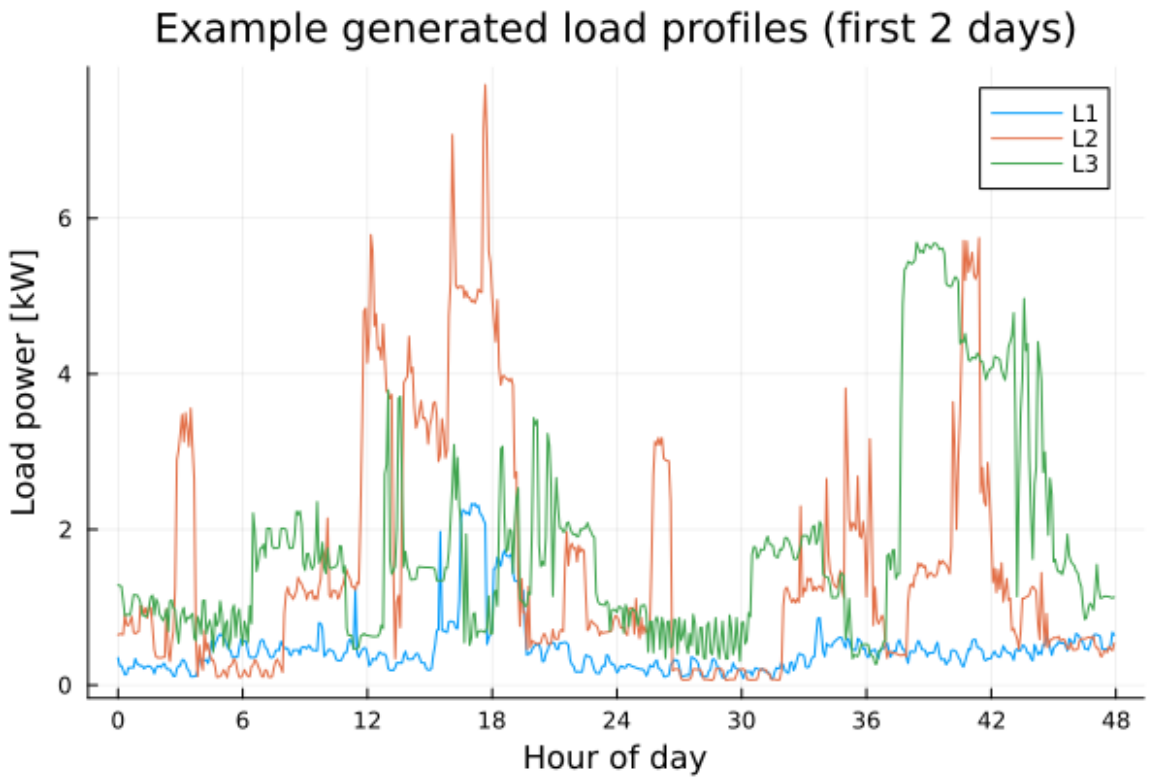


Figure 2 Example Diagram

4 Time-Series Power Flow Setup

After establishing the feeder model and processing the load profiles, yearly LoadShape objects were created in OpenDSS. The full-year time-series simulation was performed with a 5-minute time step, corresponding to 105120 simulation steps over the year.

The simulation workflow was deliberately built in stages:

1. a static snapshot model for network validation,
2. a short-duration week-1 simulation for debugging and verification,
3. a full-year simulation,
4. a dedicated yearly unbalance scan, and

5. a worst-case snapshot re-solve.

This staged workflow was beneficial for two reasons. First, it reduced debugging complexity, since topology, profile length consistency, and voltage performance could be checked incrementally. Second, it improved confidence in the final results, because the full-year study was only attempted after the static and weekly runs were verified.

All yearly time steps converged successfully. This is important because a non-converged yearly simulation would cast doubt on the reported voltage minima, maxima, and unbalance indices.

5 Results and Discussion

5.1 Static and week-1 validation

The static network model was first solved successfully, confirming that the feeder topology, transformer settings, grounding arrangement, and temporary load placement were numerically valid. The corresponding bus voltages were all within the required range.

A week-1 time-series simulation was then performed. All 2016 time steps converged. The week-1 minimum voltage was approximately 0.9104 p.u., and the week-1 maximum voltage was approximately 1.0094 p.u.. This provided an important intermediate validation before launching the full-year simulation.

The week-1 feeder active power ranged from approximately 7.46 kW to 101.15 kW, showing realistic intra-week variation. This confirms that the processed load profiles were sufficiently diverse and that the network was responding dynamically to customer demand rather than exhibiting an artificially flat or uniform pattern.

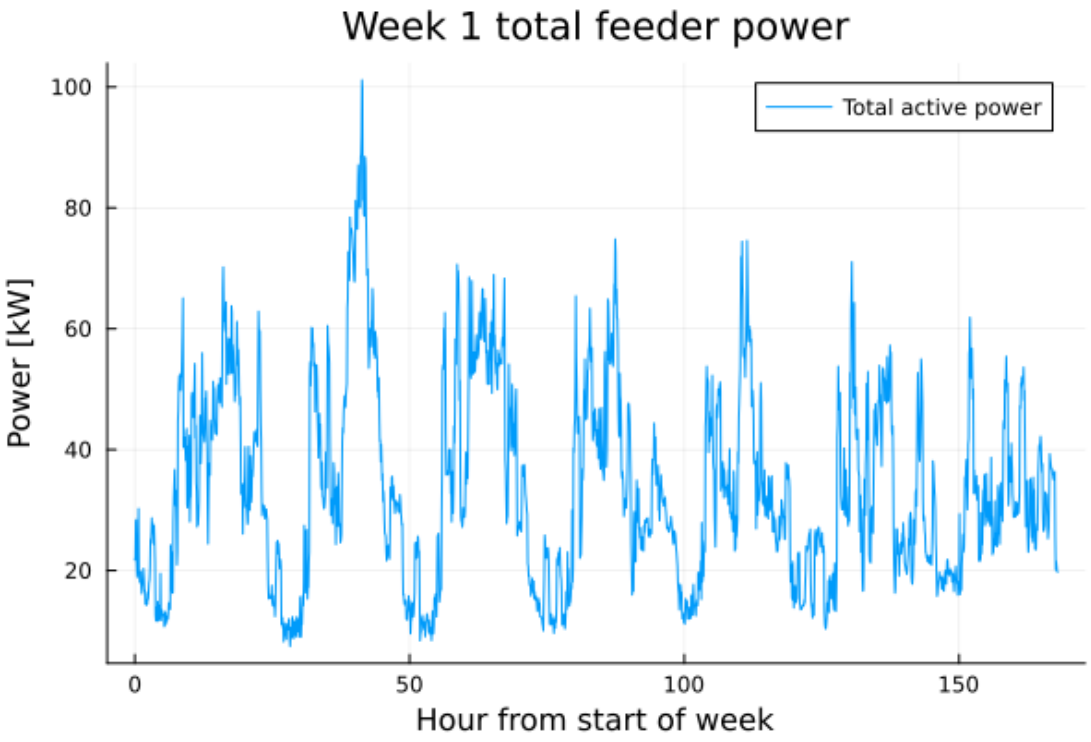


Figure 3 Week1 Total Feeder Power

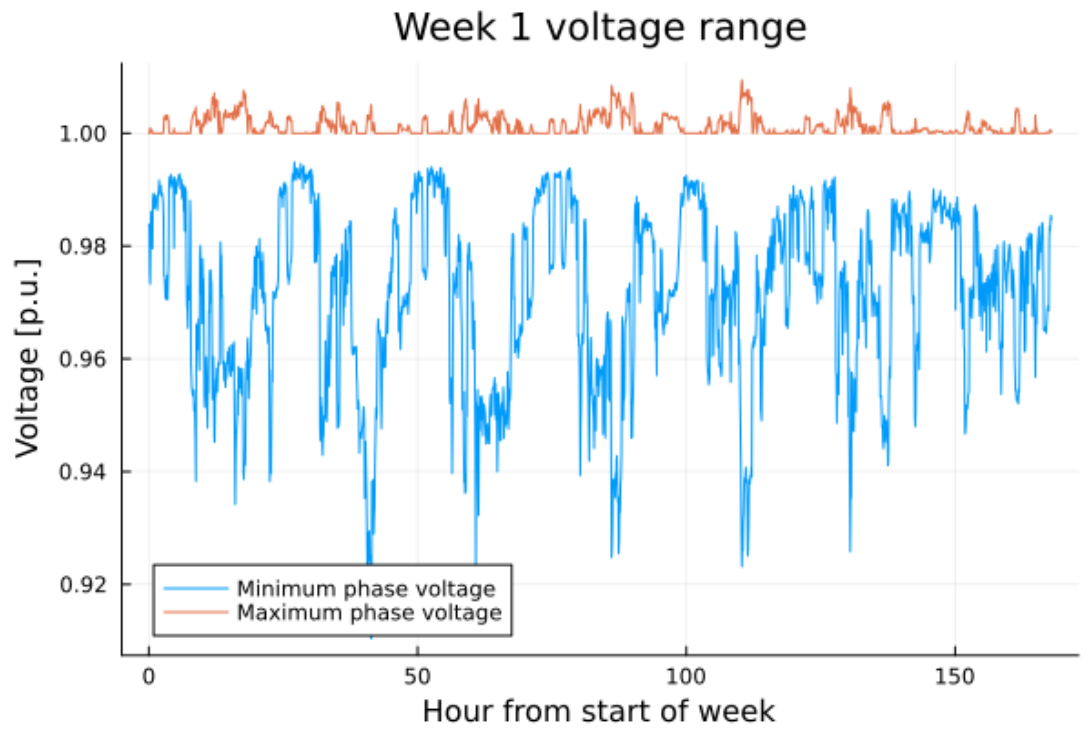


Figure 4 Week1 Voltage Range

5.2 Full-year voltage behaviour

The full-year simulation contained 105120 time steps, and all steps converged successfully. The yearly minimum voltage was:

Vmin = 0.9062809900527734 p.u.

occurring at bus b6a at:

2019-09-26 11:05:00 (+10:00)

The yearly maximum voltage was:

Vmax = 1.0114942157874707 p.u.

Therefore, the network remained fully within the assignment voltage limit of 0.9–1.1 p.u. throughout the year.

This result is significant because it demonstrates that the final feeder design achieves a realistic but still acceptable voltage profile under full-year varying demand. In other words, the network is not artificially over-designed to keep voltages unrealistically flat, but neither is it so weak that the voltage constraint is violated.

The lowest voltage occurred at b6a, which is consistent with engineering expectation. Since b6a is one of the most electrically remote buses, the current flowing through the upstream line sections accumulates a voltage drop before reaching this node. This behaviour is physically meaningful and supports the credibility of the model.

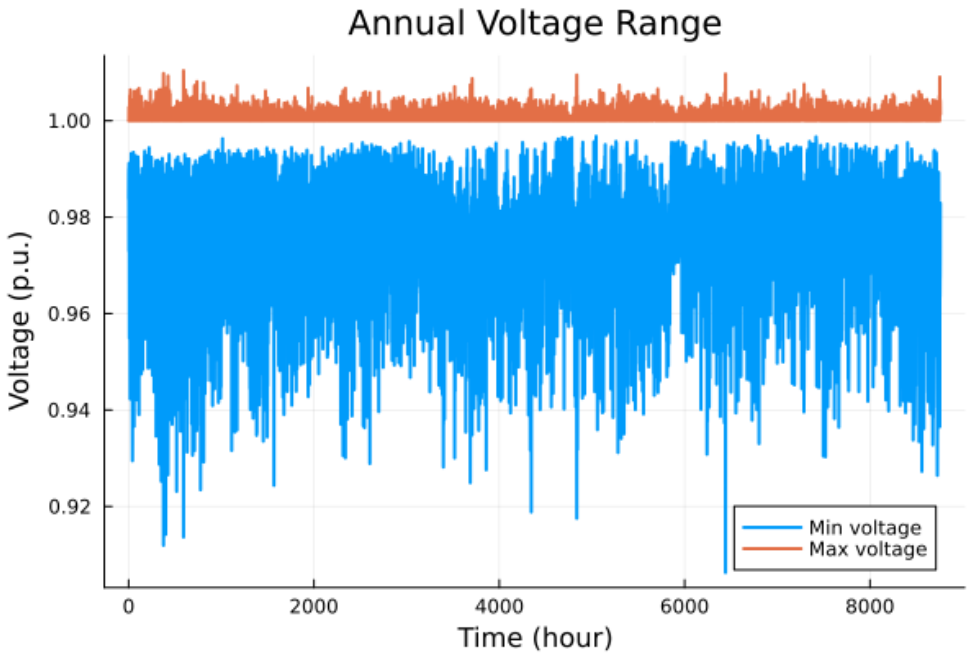


Figure 5 Annual Voltage Range

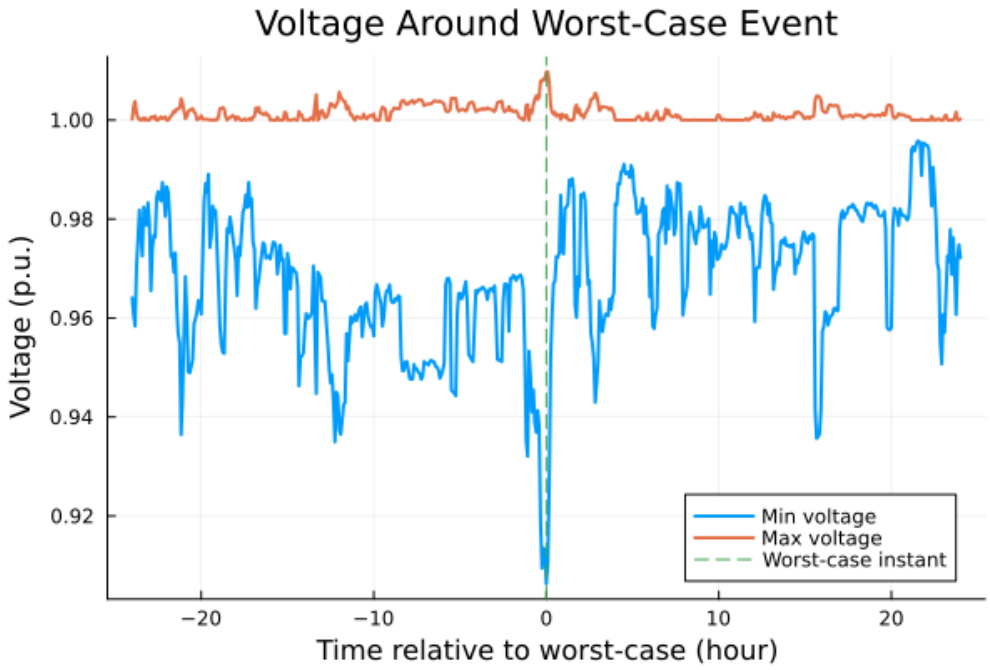


Figure 6 Voltage around Worst-case Event

5.3 Full-year feeder active power

The annual feeder active power varied substantially over time, reflecting the diversity of residential demand patterns throughout the year. The maximum feeder active power was approximately:

Pmax = 101.148902335775 kW

which occurred at:

2019-01-02 20:25:00 (+11:00)

The feeder power profile indicates that the network is neither extremely lightly loaded nor unrealistically saturated. This is desirable because it means the study captures a realistic operating range in which voltage performance and phase unbalance are both observables.

A useful observation is that the worst-case voltage event does not necessarily have to occur at the absolute annual maximum feeder power. This reflects the fact that voltage behaviour depends not only on total feeder demand, but also on the spatial distribution of the load, the phase allocation, and the specific state of the remote buses at that time.

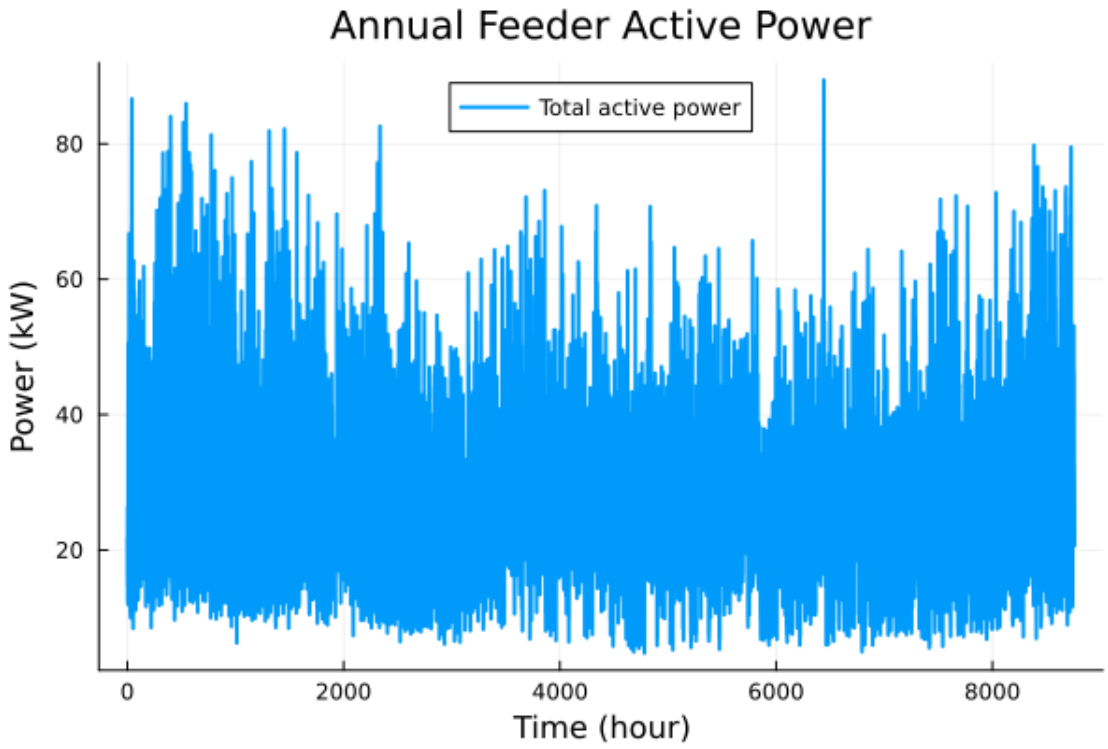


Figure 7 Annual Feeder Active Power

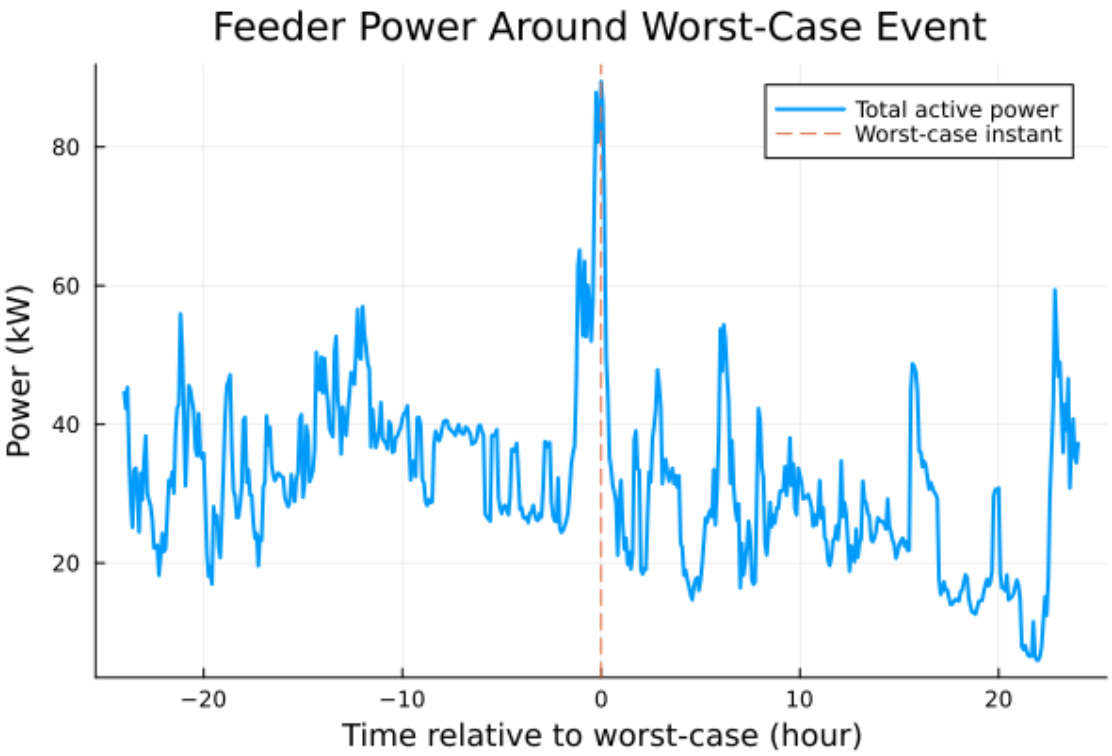


Figure 8 Feeder Power around Worst-case Event

5.4 Worst-case bus comparison

At the worst-case instant, the ten weakest buses were ranked according to minimum phase voltage. The weakest bus was b6a, followed by b6, b5, b4a, and b4. This ordering is highly informative because it demonstrates a clear electrical distance effect.

At the same worst-case instant, the key-bus phase voltages were approximately:

- b4: Va = 0.914149, Vb = 0.927489, Vc = 1.00294
- b4a: Va = 0.910666, Vb = 0.924266, Vc = 1.00287
- b5: Va = 0.908750, Vb = 0.921047, Vc = 1.00260
- b6: Va = 0.907236, Vb = 0.918358, Vc = 1.00234
- b6a: Va = 0.906284, Vb = 0.916684, Vc = 1.00217

These values show two important behaviours. First, the lowest voltage becomes progressively lower from b4 to b6a, confirming the expected voltage drop accumulation towards the feeder end. Second, phase A is consistently the weakest phase among the focused buses, while phase C remains close to 1.0 p.u. This clearly demonstrates phase-dependent voltage unbalance rather than a purely uniform three-phase drop.

If only the minimum voltage value had been reported without a bus-by-bus phase comparison, an important part of the engineering story would have been lost. The bus comparison plot makes it possible to see both the location effect and the phase effect simultaneously.

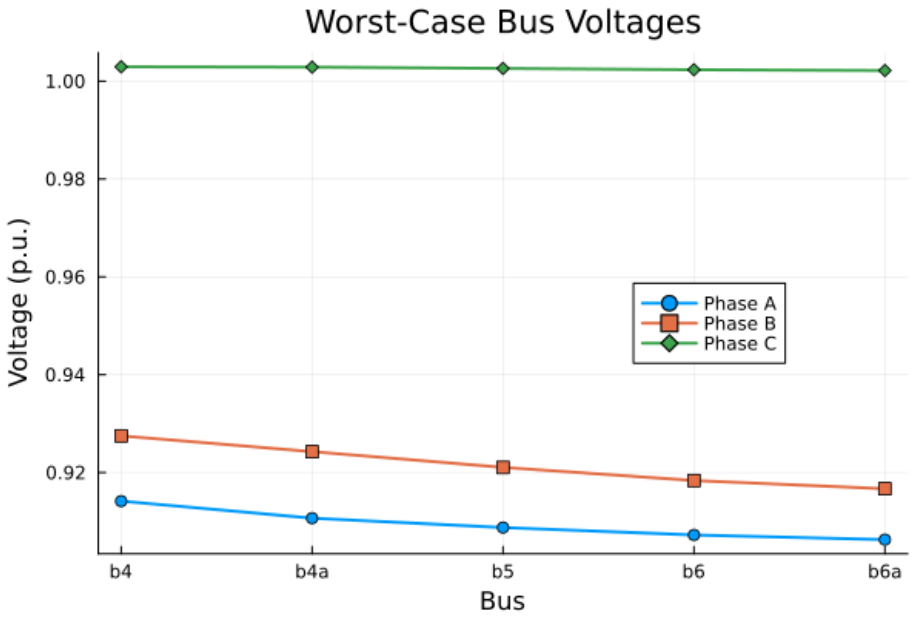


Figure 9 Worst-case Bus Voltage at Key Buses

5.5 Voltage unbalance analysis

In addition to voltage drop, the assignment also requires an evaluation of voltage unbalance. In this study, both PVUR and VUF were calculated for the full year.

The worst yearly values were:

PVURmax = 6.419462797671052% and **VUFmax = 3.009324411153066%**

Both occurred at bus b6a at the same worst-case timestamp:

2019-09-26 11:05:00 (+10:00)

This result is important because it shows that the worst voltage drop and the worst voltage unbalance are spatially and temporally aligned in the most remote part of the network. That is a physically meaningful outcome: the buses most vulnerable to voltage drop are also often the most sensitive to asymmetric phase loading.

The likely cause of this imbalance is the use of many single-phase residential loads whose magnitudes vary independently over time. Even though the feeder was designed with approximately balanced load counts per phase, equal numbers of loads do not guarantee equal time-varying power on each phase. Therefore, meaningful voltage unbalance still emerges.

This is a strong example of why merely counting loads per phase is not enough in LV feeder design. A network may look balanced on paper in terms of load count, but still behave unbalanced in operation due to profile diversity and dynamic demand variation.

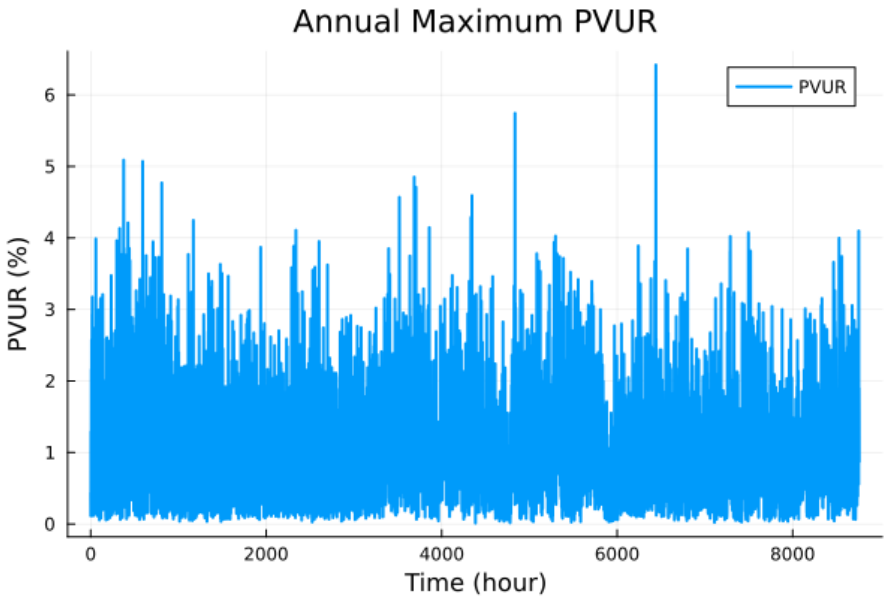


Figure 10 Annual PVUR

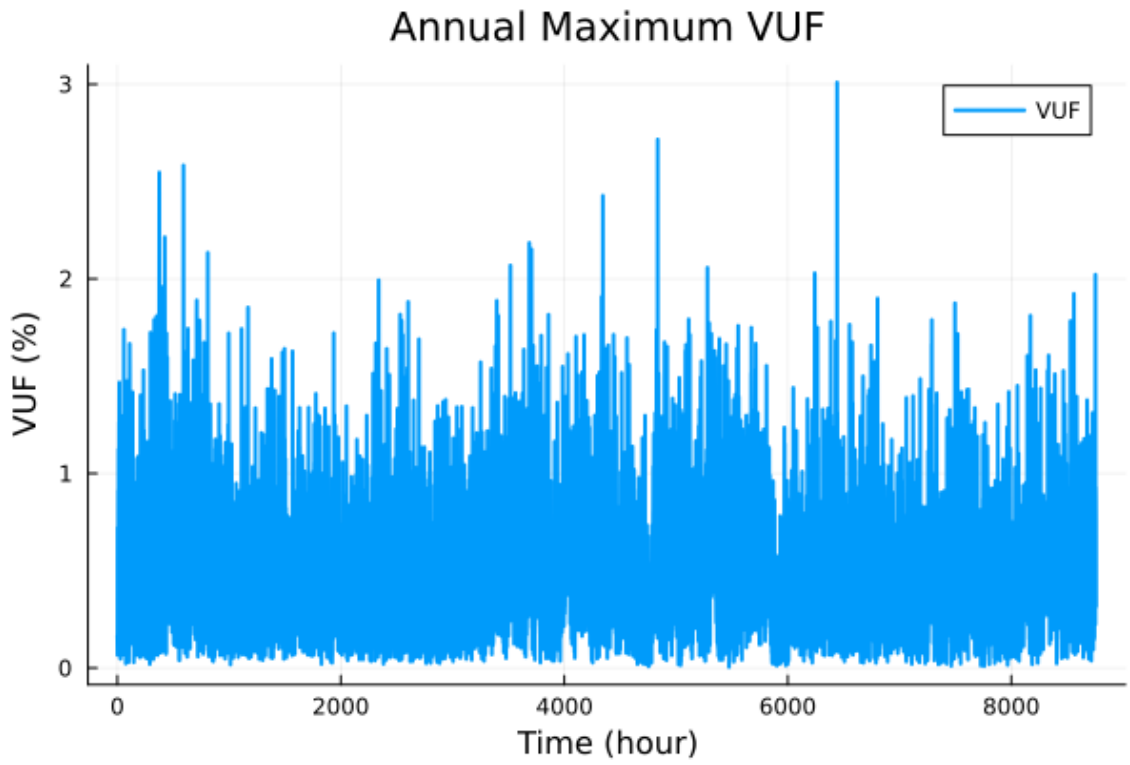


Figure 11 Annual VUF

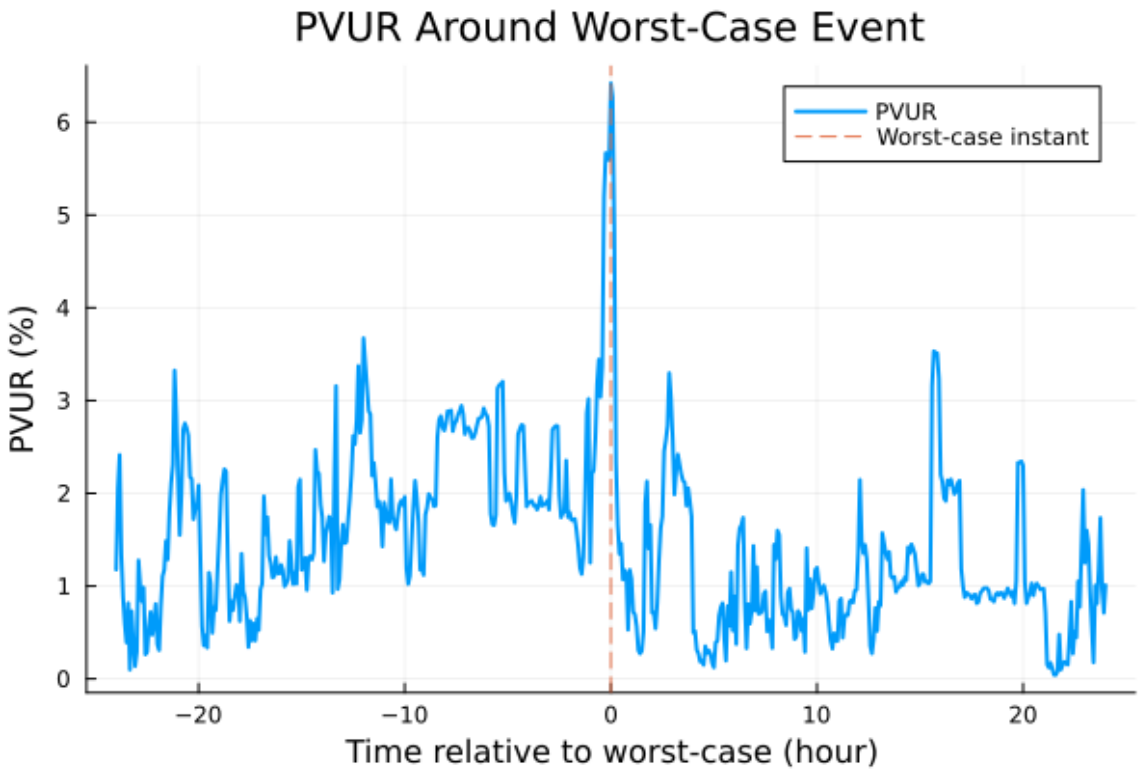


Figure 12 PVUR around Worst-case Event

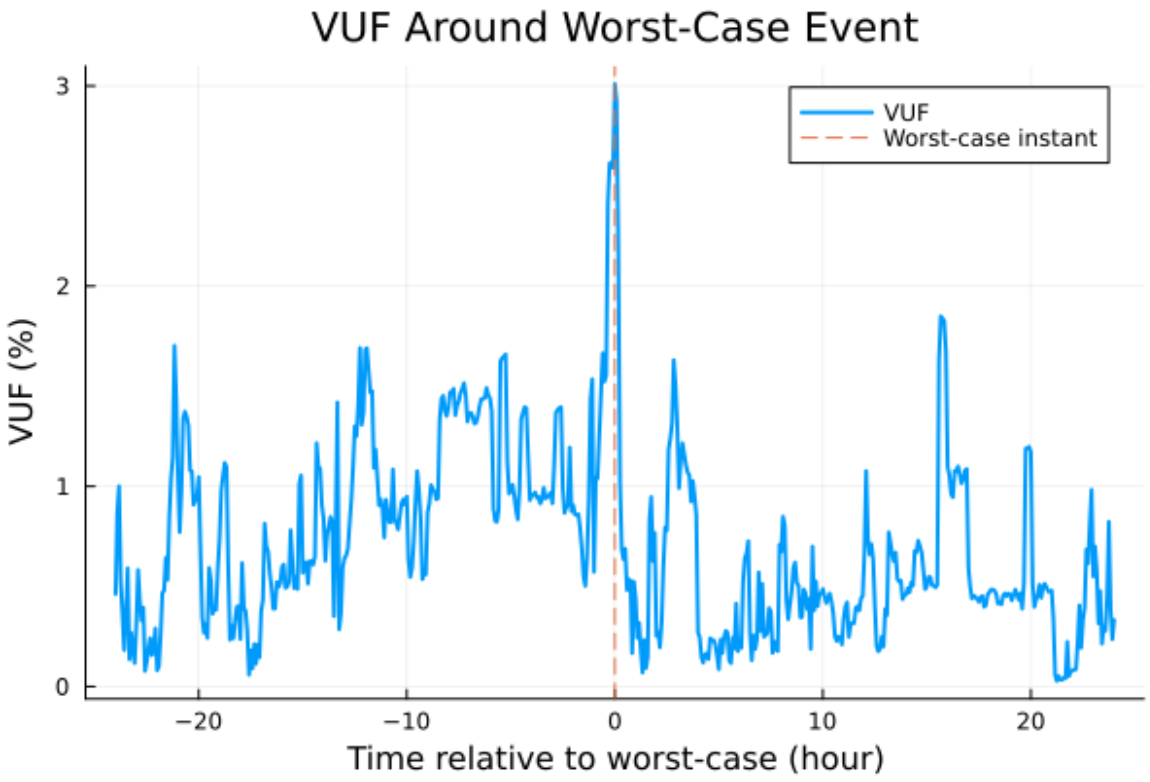


Figure 13 VUF around Worst-case Event

6 Critical Discussion

6.1 Load Profile Selection and Scaling Strategy

Although the provided dataset contains a large number of smart meter profiles, only nine representative profiles were selected in this study. These profiles were then expanded to construct 36 single-phase residential loads through scaling and allocation across different buses and phases.

This approach reflects a deliberate modelling decision rather than a data limitation. By reusing a limited number of real profiles, the model retains realistic temporal demand characteristics while maintaining computational tractability and interpretability.

However, this strategy introduces partial correlation between loads, as multiple customers share identical base profiles. This leads to an overestimation of load coincidence, where multiple loads peak simultaneously, potentially exaggerating voltage drop.

This effect was clearly observed in the initial model, where severe undervoltage (down to 0.737 p.u.) and convergence issues occurred. Therefore, while the approach is efficient and widely used, it represents a conservative scenario in terms of network stress.

This issue became particularly evident in the initial modelling stage, where excessive load coincidence led to severe undervoltage (as low as 0.737 p.u.) and even convergence difficulties. This highlighted that the naive reuse of identical profiles without sufficient diversity can significantly distort network behaviour.

6.2 Model Development and Iterative Refinement

The network model was not obtained in a single step but through an iterative refinement process based on simulation feedback and engineering reasoning.

Further improvements were made by replacing the original conductor with a lower-resistance conductor (Neptune), which reduced feeder impedance and improved voltage profiles along the network.

Increasing the transformer capacity from 300 kVA to 500 kVA was also tested. However, this change did not significantly improve the worst-case voltage, indicating that the primary limiting factor was feeder voltage drop rather than transformer loading.

The total feeder length was then reduced to approximately 0.56 km, which lies within the required range of 0.3–1.5 km. This adjustment directly reduced cumulative voltage drop and improved system performance.

6.3 Worst-Case Identification and Local Analysis

The minimum voltage was found to occur at bus b6a, the electrically most remote node in the feeder. Instead of applying arbitrary adjustments, a targeted analysis was conducted at this specific time instant.

This analysis revealed that the undervoltage issue was not uniformly distributed across phases. Instead, phase A was significantly more heavily loaded than phases B and C.

This highlights the importance of location-specific and phase-specific analysis in LV networks, where single-phase loads dominate system behaviour.

6.4 Phase Rebalancing and Final Adjustment

To address the identified imbalance, selected loads were reassigned from phase A to phase B. This phase rebalancing reduced the load asymmetry and improved the minimum voltage to within acceptable limits.

This adjustment demonstrates that phase allocation plays a critical role in voltage performance, even when the number of loads per phase is initially balanced.

If this rebalancing step had not been performed, the network would likely have continued to violate voltage constraints at the feeder end.

6.5 Voltage and Unbalance Performance

After all refinements, the network successfully maintained voltage within the required range of 0.9–1.1 p.u. over the full year.

However, voltage unbalance analysis showed that the worst PVUR and VUF still occurred at the downstream feeder, indicating that load distribution remains the dominant factor affecting network performance.

This result is consistent with physical expectations, as downstream buses experience both accumulated voltage drop and greater sensitivity to phase imbalance.

7 Conclusion

This study successfully developed a realistic three-phase four-wire LV distribution network model in OpenDSS using real smart meter demand data. The network satisfied all assignment constraints in terms of topology, transformer configuration, grounding design, line length, and load numbers.

A full-year simulation with 105120 time steps was completed successfully, and all steps converged. The voltage remained within the required range of 0.9–1.1 p.u. throughout the year. The worst-case voltage was identified at bus b6a, while meaningful voltage unbalance was also observed at the remote end of the feeder.

Overall, the results demonstrate that the model is technically valid, physically interpretable, and suitable for studying the interaction between feeder location, load variation, voltage drop, and phase unbalance in a realistic LV distribution setting.

8 Appendix

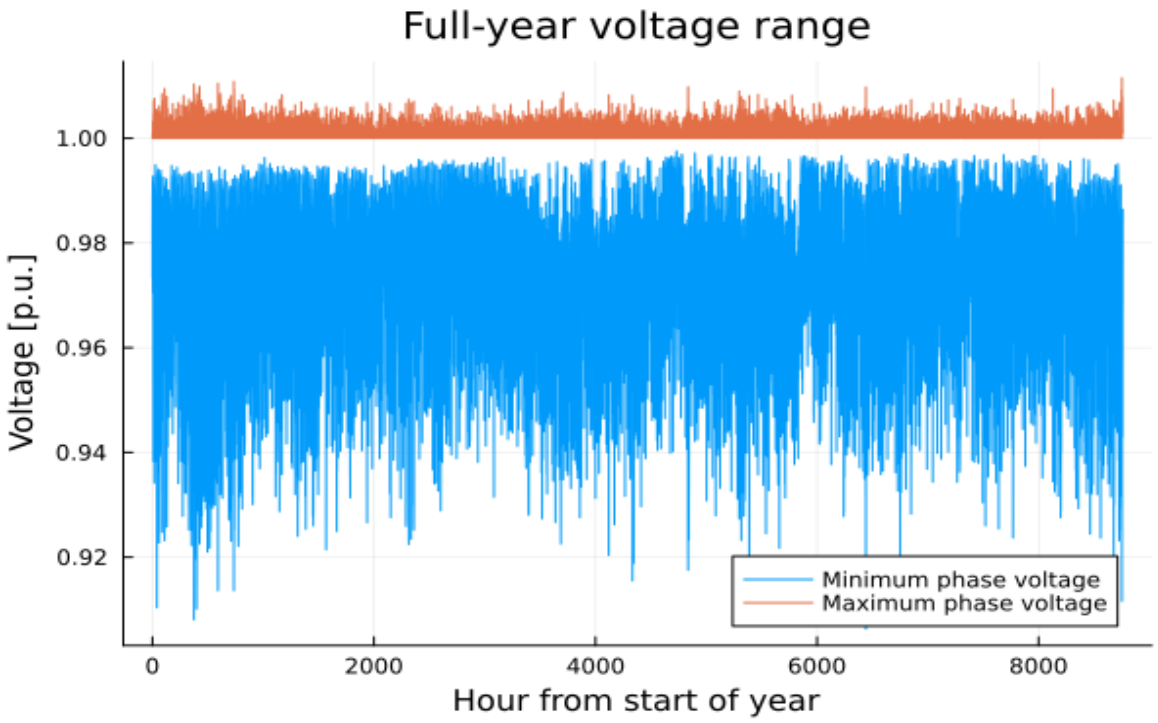


Figure 14 Original Full-year Voltage Range Plot

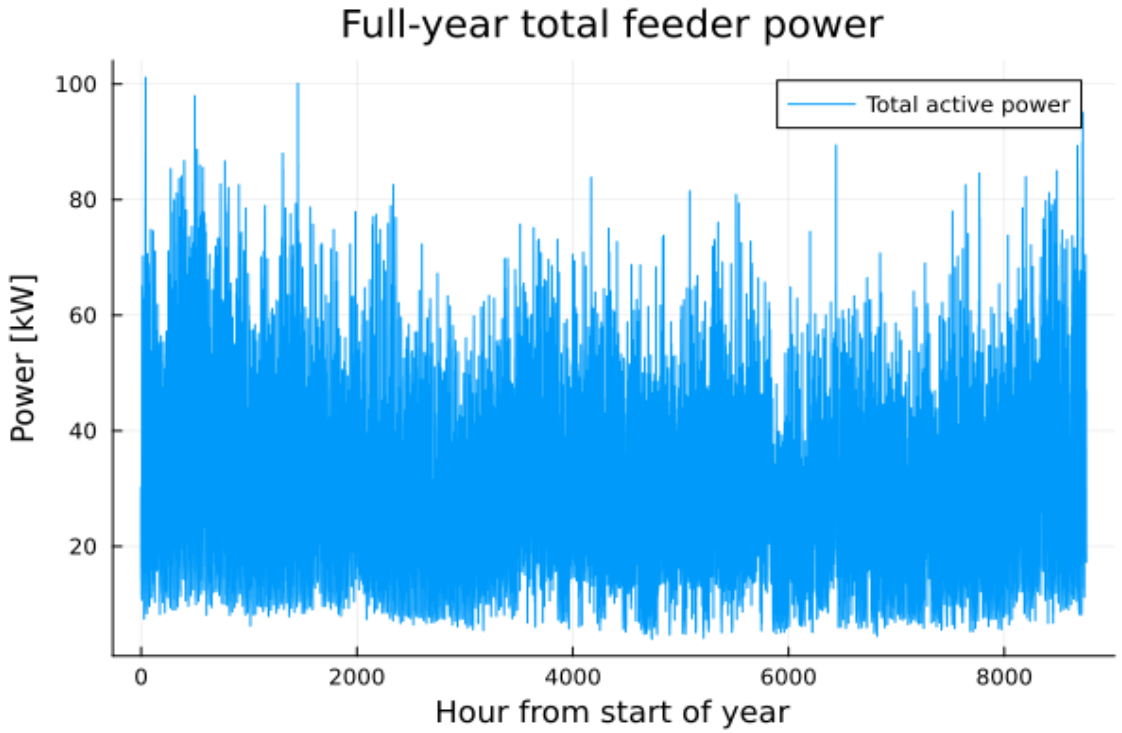


Figure 15 Original Full-year Total Feeder Power Plot

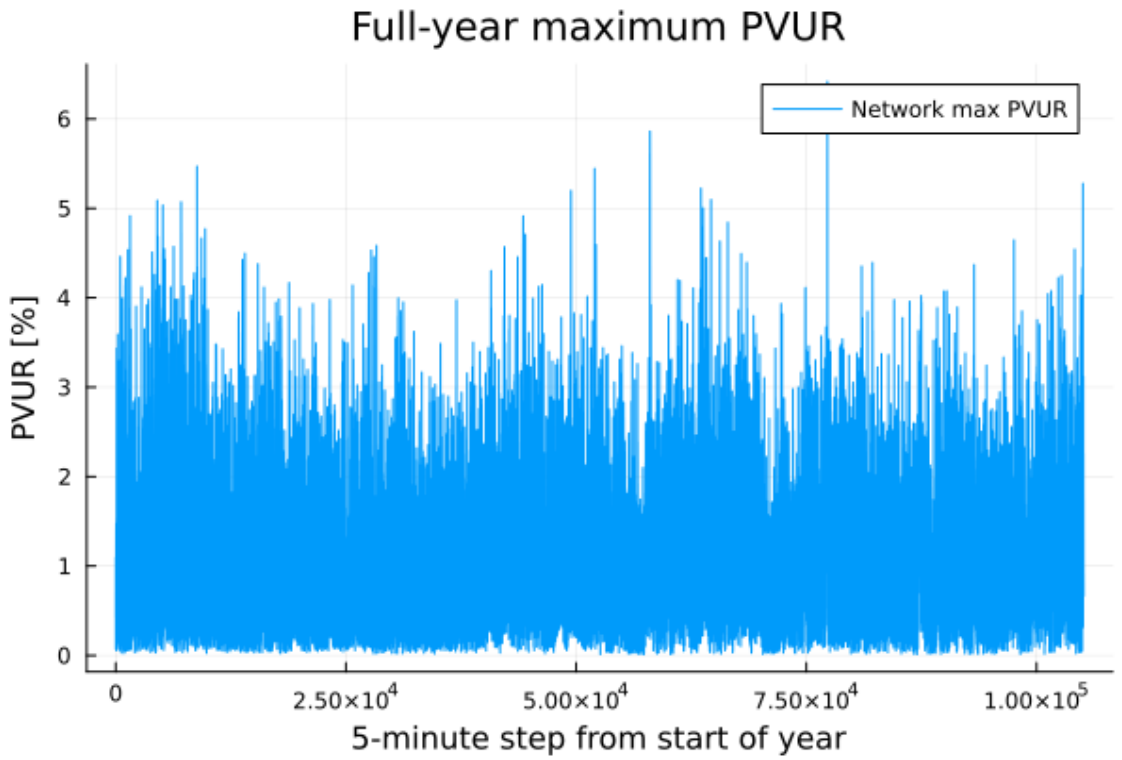


Figure 16 Original Full-year PVUR Plot

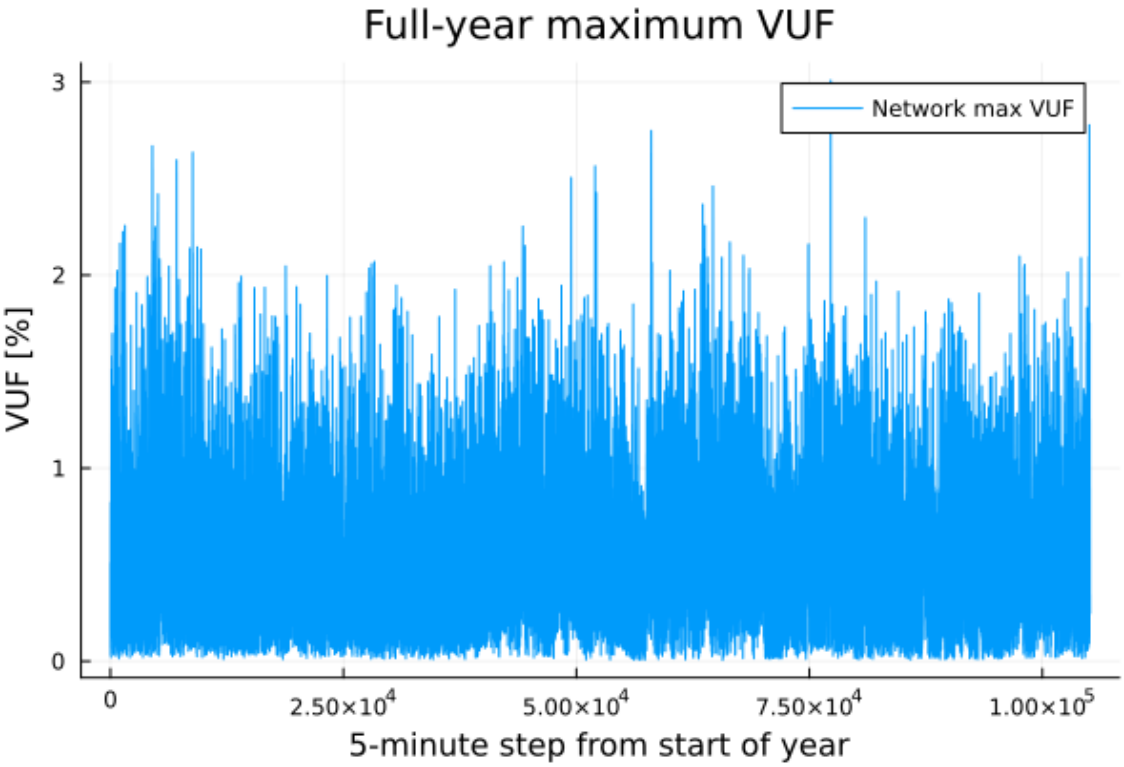


Figure 17 Original Full-year VUF Plot

bus	Va_pu	Vb_pu	Vc_pu	min_phase_v
b6a	0.906284	0.916684	1.002175	0.906284
b6	0.907236	0.918358	1.002337	0.907236
b5	0.90875	0.921047	1.0026	0.90875
b4a	0.910666	0.924266	1.002872	0.910666
b4	0.914149	0.927489	1.002938	0.914149
b3	0.92618	0.940912	1.004336	0.92618
b2a	0.94076	0.954677	1.005214	0.94076
b2	0.940914	0.954863	1.005627	0.940914
b1	0.962119	0.972023	1.007397	0.962119
tbus	0.983763	0.989782	1.009695	0.983763
sourcebus	0.999968	0.999965	0.999997	0.999965

Figure 18 Bus Voltage Worst-case Table

Github URL: https://github.sydney.edu.au/GregorVerbicGroup/ELEC5208_Task1_Group36

Task allocation:

Holistic issues such as data selection and topology design: All group members

Code: Haotian Guo

The Code Modification and Improvement: Zhengsheng Zhao

Results recording: Haotian Guo & Zhengsheng Zhao

Inspection of all work & Upload documents: Yiming Qin & Yang Jin

Report writing: Yiming Qin & Yang Jin